

Geometric Visualization of the $3n + 5$ Problem: *Voronoi Tessellation and Six Stable Basins of Attraction in Binary Logarithmic Spiral Space*

Hiroshi Harada — May 15, 2026

Abstract

This study maps the orbital structure of the $3n + 5$ dynamical system into a binary logarithmic spiral space that preserves 2-adic properties, and visualizes its behavior using Voronoi tessellation.

$$T(n) = \begin{cases} n/2 & (n \text{ even}) \\ 3n + 5 & (n \text{ odd}) \end{cases}$$

In the $3n + 5$ map, the addition term “+5” at odd steps introduces significant orbital diversity. Computational analysis reveals six distinct periodic orbits (attractor cycles). The visualization shows that natural numbers (1 to 20000) are cleanly divided into six Voronoi basins of attraction, each exhibiting a characteristic geometric skeleton (e.g., the $19 - 31 - 49$ triangular loop).

1. Introduction

While the Collatz map ($3n + 1$) is conjectured to have a single attractor (1), the $3n + 5$ map exhibits remarkable orbital diversity and instability due to the replacement of the addition term with 5. This study investigates the geometric structure of these orbits under the hypothesis that the instability arises from interference between 2-adic and 3-adic behavior.

2. Methodology

2.1 Binary Logarithmic Spiral Mapping

Natural numbers n are mapped to the polar coordinates: $r = \log_2(n)$, $\theta = 2\pi\{r\}$

This mapping preserves the following properties and faithfully reflects the geometric arrangement of natural numbers:

- 2-adic valuation (hierarchy of powers of 2)
- Logarithmic scaling
- Phase rotation with period 1

2.2 Six Periodic Orbits

The odd-step transformation $n \rightarrow 3n + 5$ acts 3-adically as a combination of:

- a one-digit left shift (multiplication by 3),
- a “+1” carry to the upper digit (from $5 = 3 + 2$),
- and a “+2” addition to the lower digit.

This dual structure causes branching in the orbits. Computational analysis confirms six periodic orbits: 1, 5, 19, 23, 187, 347

2.3 Voronoi Tessellation

Voronoi tessellation is applied to the mapped point set. Color-coding each point by its eventual periodic orbit reveals the geometric boundaries of the basins of attraction.

3. Results

3.1 Six Basins of Attraction

Natural numbers fall into the following six basins: 1, 5, 19, 23, 187, 347

Voronoi tessellation separates these basins clearly, producing a six-color tiling of the space (Figure 1).

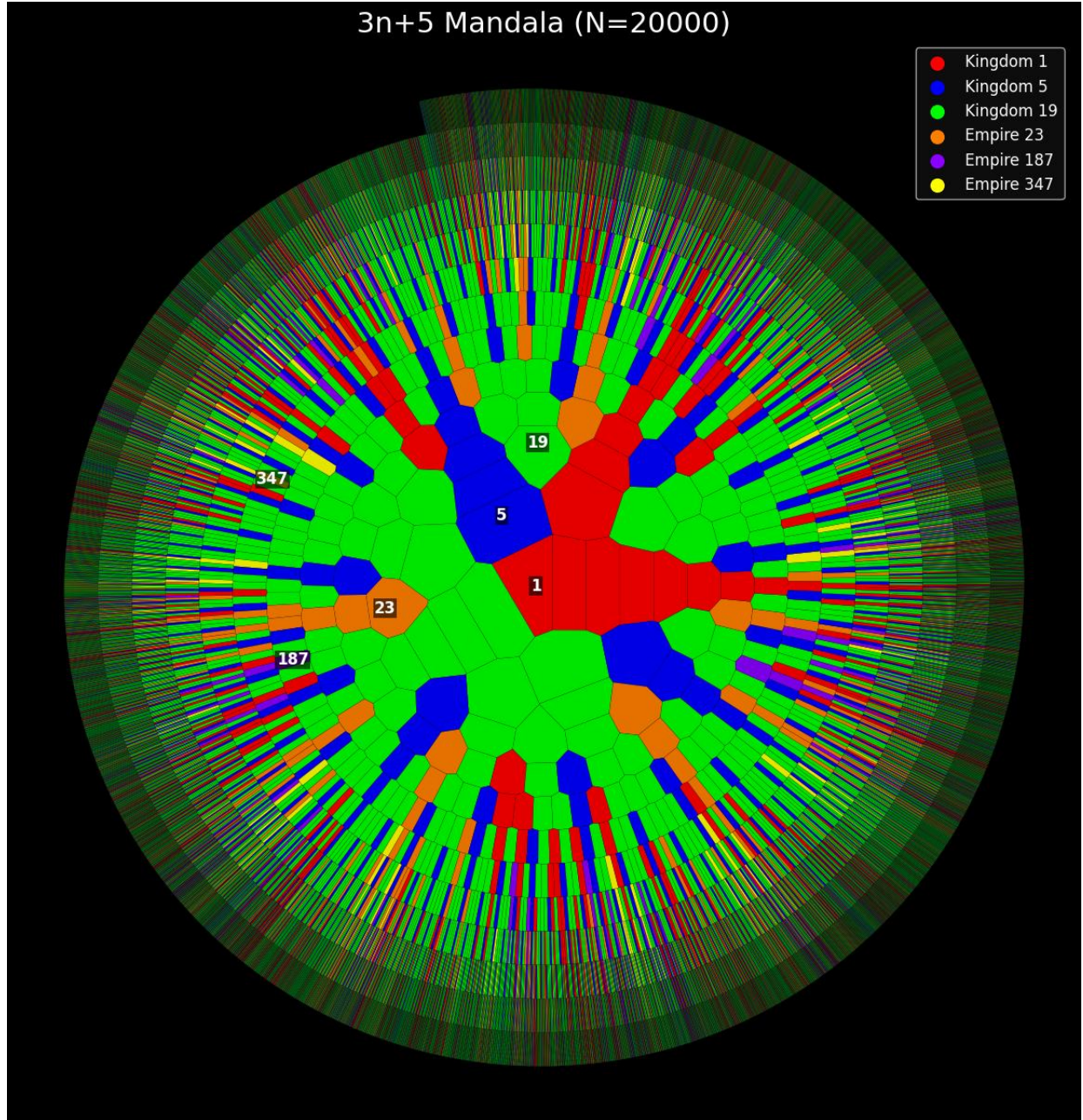


Figure 1. Voronoi tessellation of $3n + 5$ orbits in binary logarithmic spiral space. Orbits converging to the 1, 5, 19, 23, 187, or 347 periodic orbits tile the entire space as six distinct color basins.

3.2 The 19-Series Loop

The 19-series basin contains an almost equilateral triangular skeleton formed by:

$$19, 31, 49$$

This arises from a phase difference of approximately 120° among:

$$\log_2(19), \log_2(31), \log_2(49)$$

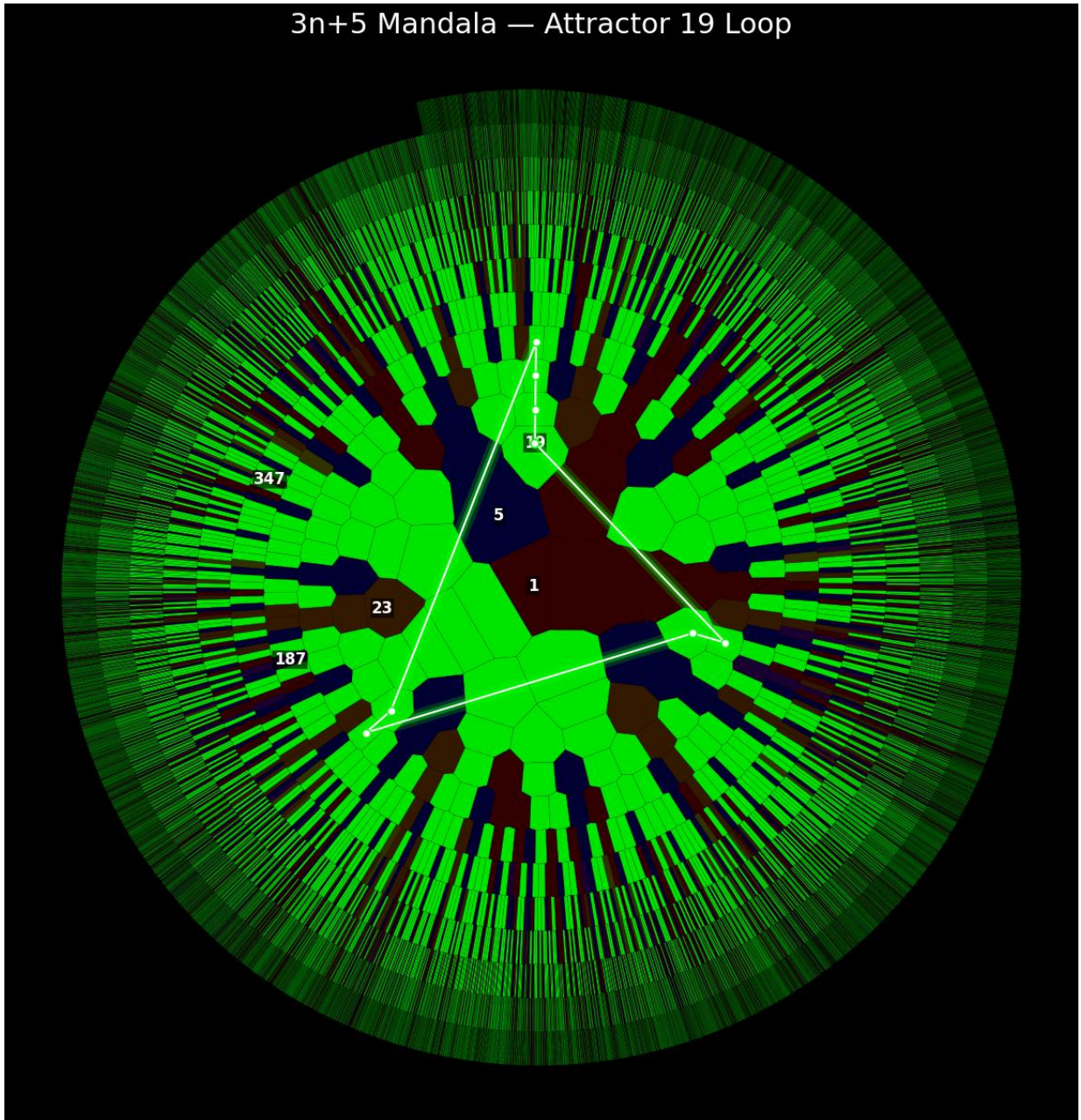


Figure 2. Geometric skeleton in the 19-series.

The 19–31–49 triangular structure formed by a phase difference of $\approx 120^\circ$.

3.3 The 23-Series Loop

Similarly, the 23-series basin exhibits a triangular structure formed by:

$$23, 29, 37$$

with phase differences of roughly 120° among:

$$\log_2(23), \log_2(29), \log_2(37)$$

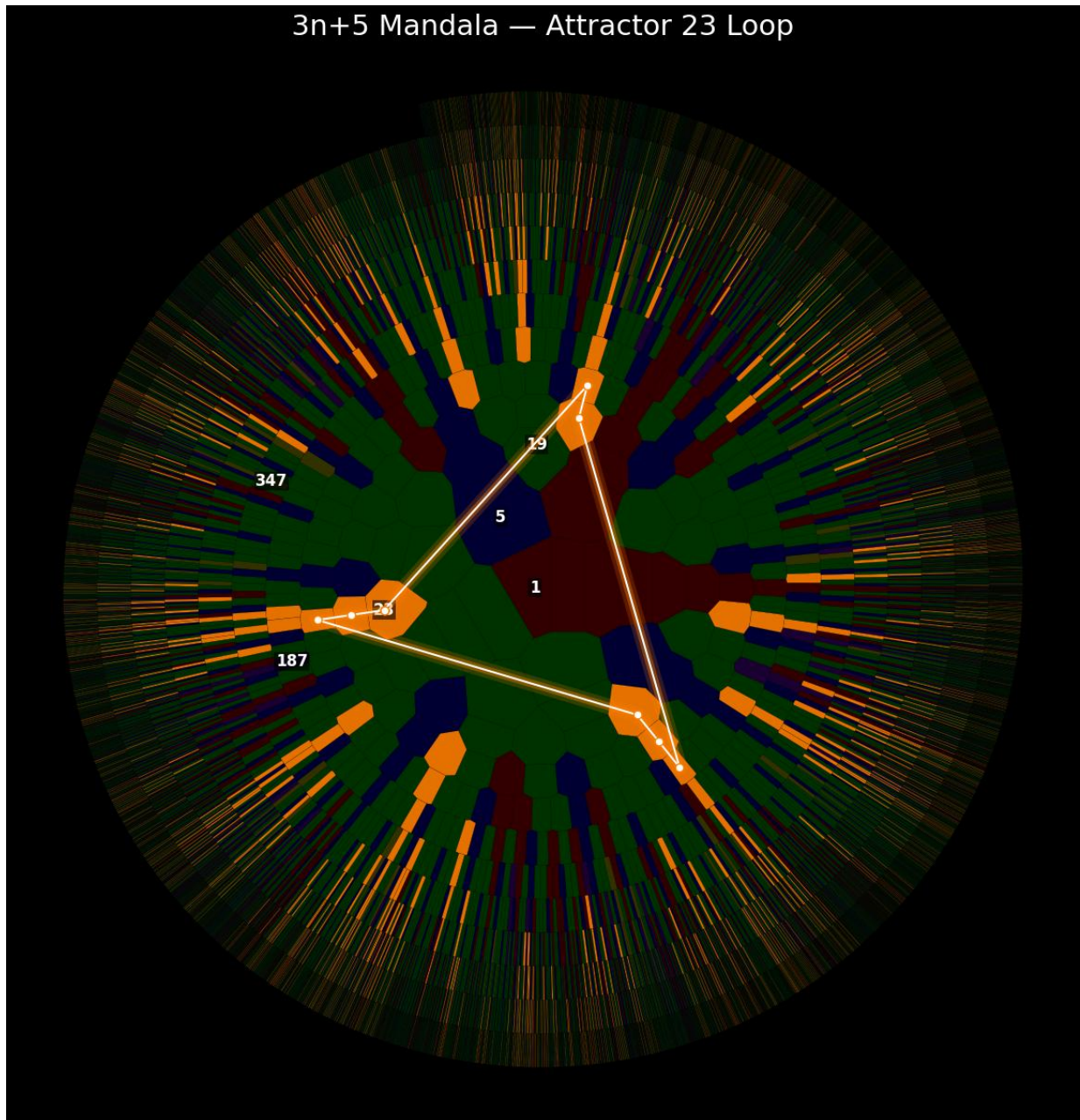


Figure 3. Geometric skeleton in the 23-series.

The 23–29–37 triangular structure formed by a phase difference of $\approx 120^\circ$.

3.4 Higher-Order Loops (187 and 347 Series)

For the 187 and 347 basins, computational analysis reveals a repeating 17-step subcycle, producing a fractal-like local loop network in the Voronoi space (**Figure 4** and **5**).

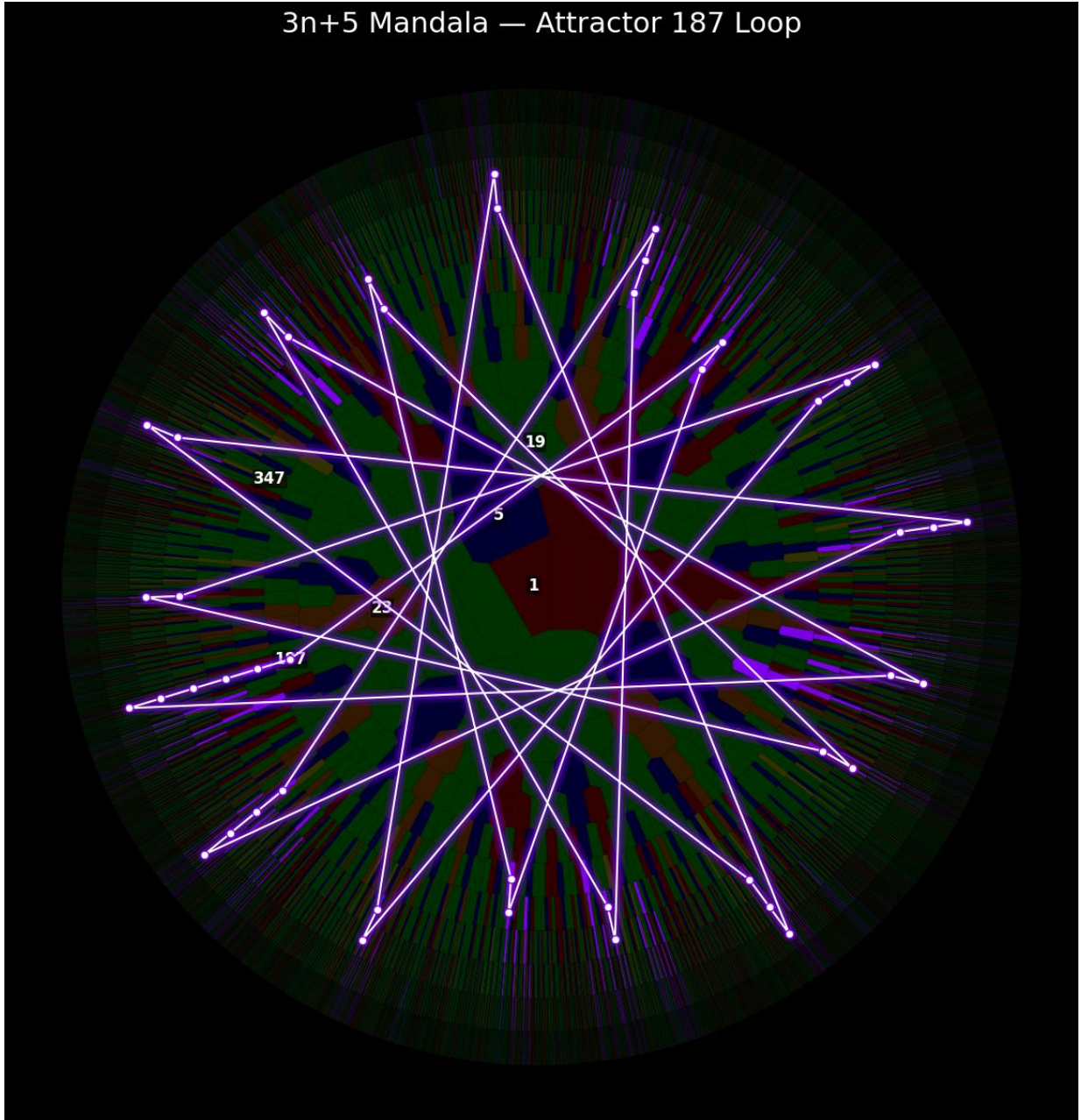


Figure 4. Geometric skeleton in the 187-series.
A repeating 17-step subcycle forms a localized loop network.

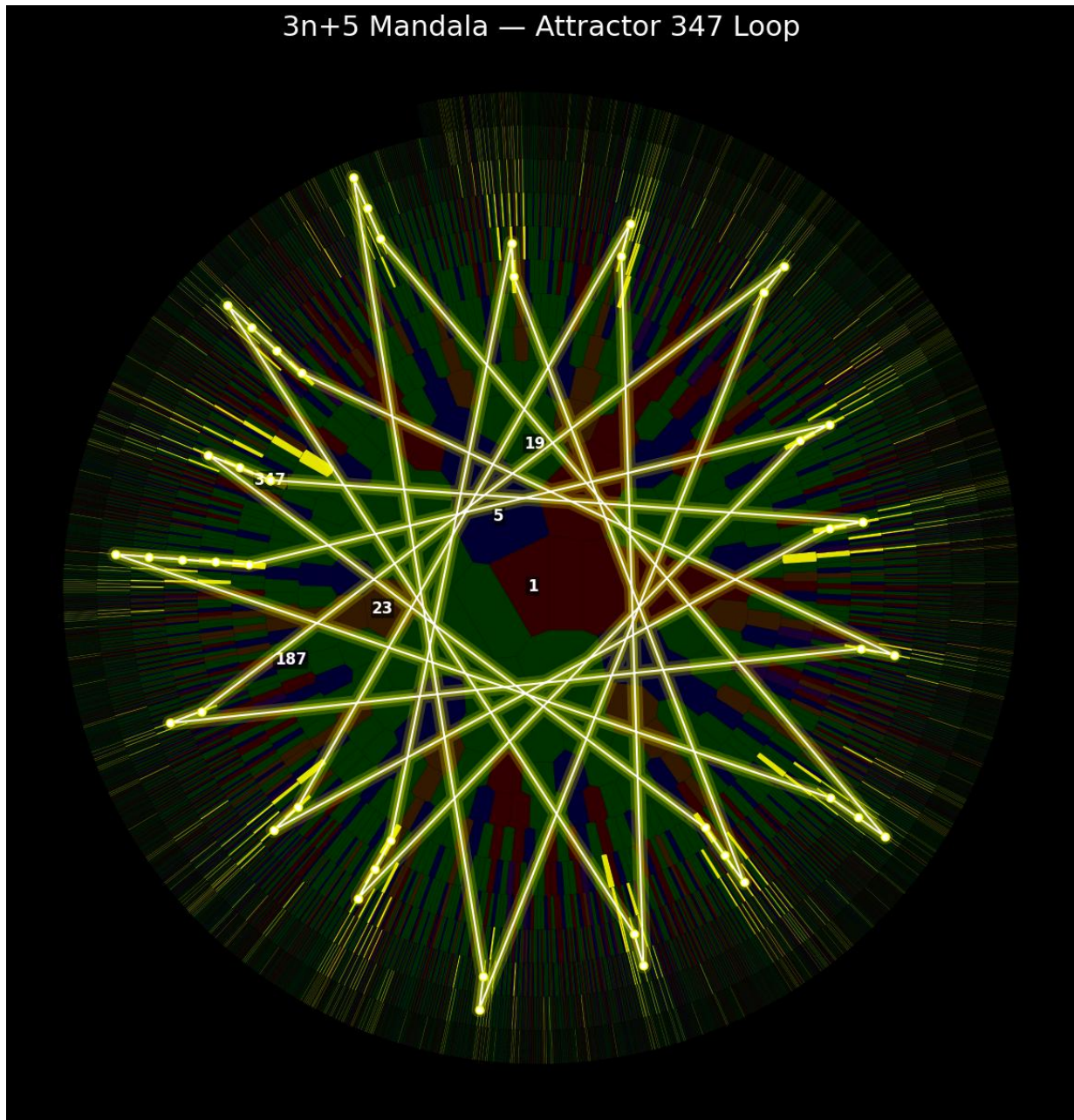


Figure 5. Geometric skeleton in the 347-series.
A repeating 17-step subcycle forms a localized loop network.

4. Conclusion

Despite its simple definition, the $3n+5$ map generates six basins of attraction and intricate geometric structures due to interference between 2-adic and 3-adic behavior.

The combination of binary logarithmic spiral mapping and Voronoi tessellation provides a powerful framework for visualizing and classifying these orbits.

Future work includes extending this approach to the broader family of $3n+k$ maps.

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